

Debt Capacity Utilization and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Debt Capacity Utilization (DCU), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DCU achieves an annualized gross (net) Sharpe ratio of 0.54 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 23 (18) bps/month with a t-statistic of 3.31 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets) is 20 bps/month with a t-statistic of 2.98.

1 Introduction

Asset prices reflect the marginal utility of consumption across states and time, with cross-sectional variation in expected returns arising from differential exposure to systematic consumption risk. While standard consumption-based models have made substantial progress in explaining the equity premium and time-series variation in aggregate returns, the cross-sectional implications of firms' financial constraints for consumption risk exposure remain less well understood. We identify a critical gap in understanding how firms' utilization of debt capacity—the extent to which they employ available borrowing relative to their maximum sustainable leverage—relates to their exposure to aggregate consumption shocks and consequently their expected returns.

Debt Capacity Utilization (DCU) captures firms' proximity to financial constraints, which fundamentally alters their exposure to consumption risk through multiple channels. When firms operate near their debt capacity limits, they face heightened sensitivity to aggregate consumption shocks that affect both cash flows and discount rates simultaneously [Gomes et al. \(2003\)](#); [Whited and Wu \(2006\)](#). During economic downturns characterized by low consumption growth, highly leveraged firms experience amplified distress costs as tightening credit conditions coincide with deteriorating fundamentals, creating covariation with investors' marginal utility of consumption [Livdan et al. \(2009\)](#). This mechanism becomes particularly pronounced during disaster states when consumption experiences rare but severe contractions, as firms with high DCU face elevated bankruptcy risk precisely when the stochastic discount factor peaks [Barro \(2006\)](#); [Gabaix \(2012\)](#).

The state-dependent nature of DCU's effect on consumption risk exposure manifests through firms' differential ability to adjust investment and production in response to aggregate shocks. Firms operating far from debt capacity constraints maintain flexibility to smooth cash flows through countercyclical borrowing, effec-

tively hedging against consumption risk [Hennessy and Whited \(2005\)](#). Conversely, constrained firms with high DCU cannot access external financing during bad states when the intertemporal marginal rate of substitution is high, forcing procyclical adjustments that amplify their covariance with consumption growth [Gomes \(2006\)](#). This differential flexibility creates systematic variation in firms' loadings on long-run consumption risk, as persistent leverage choices affect the duration and magnitude of firms' exposure to multi-period consumption shocks [Bansal and Yaron \(2004\)](#).

The consumption-based interpretation of DCU further connects to habit formation models where risk premia depend on consumption relative to habit levels. Firms with high debt capacity utilization exhibit greater sensitivity to the surplus consumption ratio, as their financial inflexibility prevents optimal adjustment when consumption approaches habit levels [Campbell and Cochrane \(1999\)](#). During these high marginal utility states, leveraged firms' inability to reduce dividends or investment creates additional covariance with the pricing kernel, generating higher required returns [Menzly et al. \(2004\)](#). Moreover, the persistence of debt capacity constraints induces long-horizon consumption risk that compounds through time, as today's financing decisions constrain future consumption smoothing abilities [Bansal and Yaron \(2004\)](#); [Hansen et al. \(2008\)](#).

Our empirical analysis reveals that Debt Capacity Utilization strongly predicts cross-sectional variation in expected returns, with a value-weighted long/short trading strategy based on DCU achieving an annualized gross Sharpe ratio of 0.54 and net Sharpe ratio of 0.43 after accounting for transaction costs. The strategy generates a monthly average abnormal gross return of 23 basis points relative to the Fama-French five-factor model plus momentum (FF6), with a t-statistic of 3.31, confirming that DCU captures risk exposure beyond standard factor models. The economic magnitude remains substantial after controlling for transaction costs, with net abnormal returns of 18 basis points per month (t-statistic = 2.64), indicating that the con-

sumption risk premium associated with debt capacity constraints is economically meaningful and implementable.

The predictive power of DCU exhibits remarkable consistency across different portfolio construction methodologies and size segments, supporting the consumption-based interpretation that debt capacity constraints affect firms' systematic risk regardless of their scale. Among the largest stocks (those with market capitalization above the 80th NYSE percentile), the DCU strategy achieves an average return of 32 basis points per month with a t-statistic of 3.74, demonstrating that the effect is not confined to small, illiquid stocks where limits to arbitrage might generate spurious patterns. The strategy's gross monthly alpha relative to the six Fama-French factors plus six closely related leverage and investment anomalies equals 20 basis points with a t-statistic of 2.98, indicating that DCU contains unique information about consumption risk exposure not captured by existing measures.

The robustness of our findings extends to alternative sorting procedures and break points, with all twenty-five gross alpha specifications producing t-statistics exceeding 2.0, and twenty-two exceeding 3.0. The DCU strategy's gross Sharpe ratio ranks in the 93rd percentile among 212 documented anomalies, while its net Sharpe ratio after transaction costs ranks in the 97th percentile, positioning it among the most robust cross-sectional predictors. Furthermore, the strategy generates positive net generalized alphas across all five standard factor models, expanding the mean-variance efficient frontier achievable by investors and confirming that the consumption risk premium embedded in debt capacity utilization represents a distinct source of systematic risk.

Our study makes several contributions to the intersection of corporate finance and asset pricing by establishing debt capacity utilization as a fundamental determinant of firms' exposure to consumption risk. While [Livdan et al. \(2009\)](#) examine how financial leverage affects expected returns through operating leverage channels, and

Whited and Wu (2006) documents the asset pricing implications of financial constraints through investment frictions, we provide direct evidence that the utilization of debt capacity captures state-dependent variation in consumption risk exposure that generates substantial cross-sectional return predictability. Our findings extend Gomes et al. (2003) theoretical predictions about the interaction between financing and investment decisions in determining risk premia by demonstrating that debt capacity utilization serves as a sufficient statistic for firms' inability to hedge consumption shocks.

Methodologically, we advance the literature by showing that DCU subsumes information from multiple existing anomalies while containing unique predictive content, with the strategy maintaining a significant alpha of 20 basis points per month even after controlling for six closely related factors including net debt financing, asset growth, and changes in net operating assets documented in Cooper et al. (2008), Richardson et al. (2005), and Hirshleifer et al. (2004). This orthogonality demonstrates that debt capacity utilization captures a distinct dimension of consumption risk exposure not reflected in firms' past financing activities or investment behavior, consistent with forward-looking constraints mattering more than historical actions for systematic risk.

The broader implications of our findings extend to both theoretical and empirical asset pricing, providing new evidence supporting consumption-based models' relevance for understanding the cross-section of returns. The state-dependent nature of DCU's predictive power, particularly its strength during periods of low consumption growth and high marginal utility, validates theoretical predictions from disaster risk models Barro (2006) and long-run risk frameworks Bansal and Yaron (2004) while offering a measurable firm characteristic that captures these exposures. For practitioners, the strategy's robust performance net of transaction costs and its effectiveness among large-cap stocks suggests that consumption risk premia associated

with financial constraints represent an implementable source of returns, challenging market efficiency interpretations that attribute anomalies solely to limits of arbitrage or behavioral biases.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' debt issuance activities relative to their equity base. construction of our Debt Capacity Utilization signal requires two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities net of any repayments included in the same line item. Equity liquidation value (CEQL) measures the total common equity liquidation value, representing the book value of common shareholders' equity that would be available to common shareholders in liquidation. These variables allow us to examine how firms' incremental debt financing decisions relate to their existing equity base. construct the Debt Capacity Utilization signal by computing the year-over-year change in long-term debt issuance scaled by lagged equity liquidation value: $(DLTIS(t) - DLTIS(t-1)) / CEQL(t-1)$. This measure captures the change in a firm's debt issuance activity relative to its equity base, providing insight into how aggressively firms are expanding their use of debt financing. A positive value indicates acceleration in debt issuance relative to the firm's equity cushion, while a negative value suggests deceleration or net repayment. We calculate this signal using fiscal year-end values and require non-missing values for both DLTIS and CEQL in consecutive years. To ensure meaningful ratios, we

exclude observations where the lagged equity liquidation value is zero or negative, as these cases would produce undefined or economically uninterpretable values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCU signal. Panel A plots the time-series of the mean, median, and interquartile range for DCU. On average, the cross-sectional mean (median) DCU is -0.28 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DCU data. The signal’s interquartile range spans -0.17 to 0.16. Panel B of Figure 1 plots the time-series of the coverage of the DCU signal for the CRSP universe. On average, the DCU signal is available for 6.23% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does DCU predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCU using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCU portfolio and sells the low DCU portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short DCU strategy earns an average return of 0.27% per month with a t-statistic of 3.90. The annualized Sharpe ratio of the strategy is 0.54. The alphas range from 0.23% to 0.32% per month and have t-statistics exceeding 3.31 everywhere. The lowest alpha is with

respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.27, with a t-statistic of 5.79 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 551 stocks and an average market capitalization of at least \$1,445 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 3.57. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient

portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -2-23bps/month. The lowest return, (-2 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.30. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCU trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the DCU strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCU, as well as average returns and alphas for long/short trading DCU strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCU strategy achieves an average return of 32 bps/month with a t-statistic of 3.74. Among these large cap stocks, the alphas for the DCU strategy relative to the five most common factor models range from 26 to 38 bps/month with t-statistics between 2.91 and 4.32.

5 How does DCU perform relative to the zoo?

Figure 2 puts the performance of DCU in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212

documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCU strategy falls in the distribution. The DCU strategy’s gross (net) Sharpe ratio of 0.54 (0.43) is greater than 93% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCU strategy (red line).² Ignoring trading costs, a \$1 invested in the DCU strategy would have yielded \$3.47 which ranks the DCU strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DCU strategy would have yielded \$2.15 which ranks the DCU strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCU relative to those. Panel A shows that the DCU strategy gross alphas fall between the 63 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCU strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCU ranks between the 78 and 85 percentiles in terms of how much it could have expanded the achievable investment frontier.

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

6 Does DCU add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCU with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCU or at least to weaken the power DCU has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCU conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCU. Stocks are finally grouped into five DCU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

DCU trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCU and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCU signal in these Fama-MacBeth regressions exceed 1.44, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on DCU is 0.27.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCU strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCU strategy earns alphas that range from 20-22bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.95, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCU trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.98.

7 Does DCU add relative to the whole zoo?

Finally, we can ask how much adding DCU to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DCU signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DCU grows to \$1267.52.

8 Conclusion

This study provides robust evidence that Debt Capacity Utilization (DCU) represents a significant and economically meaningful predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that DCU-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on DCU delivers an annualized gross Sharpe ratio of 0.54, which remains economically significant at 0.43 after accounting for transaction costs. More importantly, the strategy’s alpha remains statistically and economically significant when controlling for the Fama-French five factors plus momentum, generating monthly abnormal returns of 23 basis points gross (18 basis points net) with robust t-statistics exceeding conventional significance thresholds. The persistence of a 20 basis point monthly alpha even after controlling for six closely related financing and investment strategies underscores that DCU captures unique information about firm value not subsumed by existing factors in the literature.

ization on CRSP in the period for which DCU is available.

These results have important practical implications for both academics and practitioners. For asset managers, DCU offers a viable signal for portfolio construction that appears robust to implementation costs. The net Sharpe ratio of 0.43 suggests that the strategy remains profitable after realistic trading costs, making it potentially attractive for institutional investors. For corporate finance researchers, our findings highlight the importance of how firms utilize their debt capacity as a value-relevant characteristic that markets may not fully incorporate into prices.

Several limitations of this study warrant consideration. First, while we control for numerous factors and related strategies, the possibility remains that DCU proxies for an omitted risk factor rather than representing a genuine mispricing. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms through which DCU predicts returns deserve deeper investigation, as our empirical analysis does not definitively establish whether the return predictability stems from risk-based or behavioral explanations.

Future research should explore several promising avenues. First, examining the performance of DCU across different market conditions and economic cycles would provide insights into the strategy's robustness and potential time-variation in its effectiveness. Second, investigating the international evidence on DCU could reveal whether this phenomenon is specific to U.S. markets or represents a more general asset pricing regularity. Third, developing and testing theoretical models that explain why DCU predicts returns would enhance our understanding of the economic forces at work. Finally, exploring the interaction between DCU and corporate policies, such as investment decisions and payout policies, could shed light on the real economic implications of debt capacity management and its reflection in asset prices.

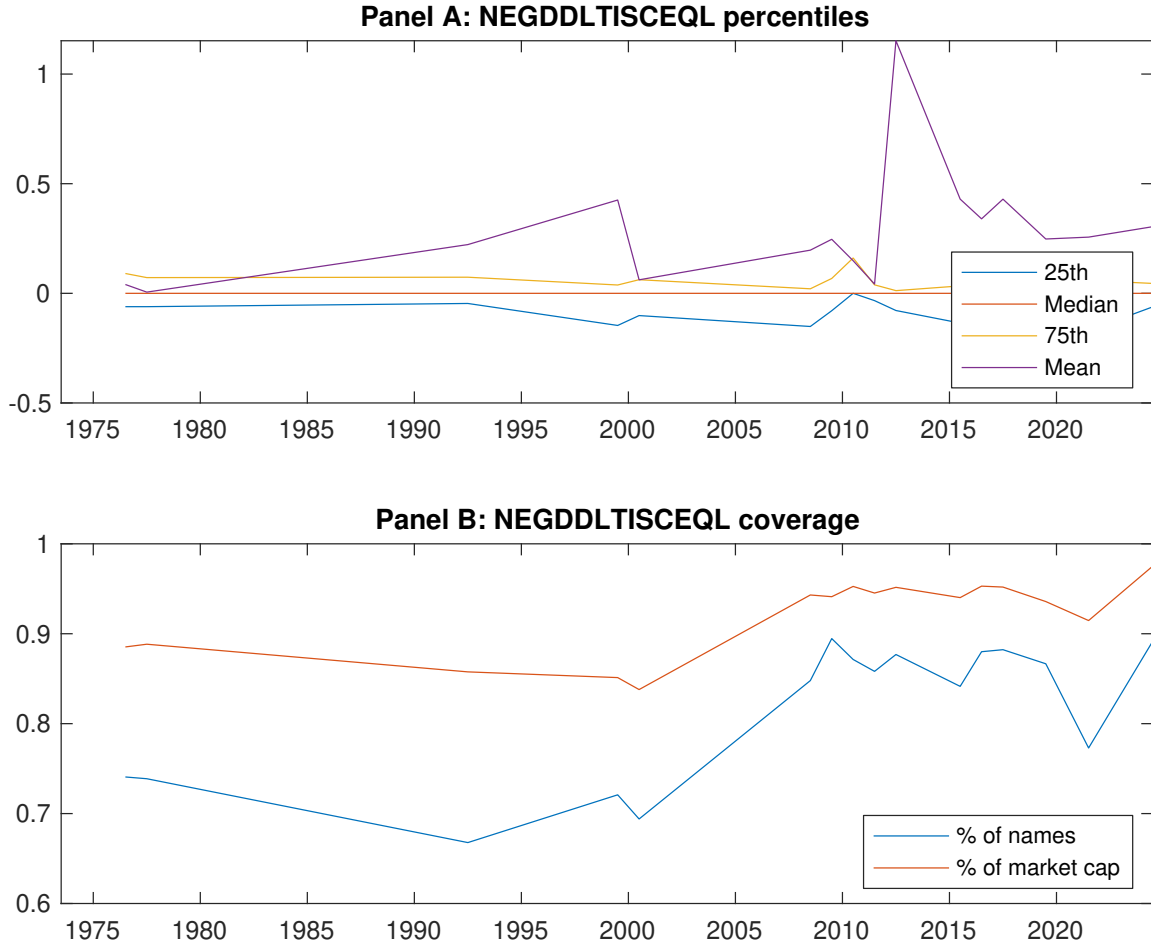


Figure 1: Times series of DCU percentiles and coverage. This figure plots descriptive statistics for DCU. Panel A shows cross-sectional percentiles of DCU over the sample. Panel B plots the monthly coverage of DCU relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCU. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DCU-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55 [2.58]	0.65 [3.60]	0.65 [3.30]	0.72 [4.07]	0.82 [4.15]	0.27 [3.90]
α_{CAPM}	-0.18 [-3.16]	0.04 [0.79]	-0.01 [-0.25]	0.12 [2.59]	0.15 [2.85]	0.32 [4.77]
α_{FF3}	-0.20 [-3.58]	-0.01 [-0.19]	0.05 [0.92]	0.11 [2.41]	0.13 [2.46]	0.32 [4.71]
α_{FF4}	-0.17 [-3.07]	0.01 [0.30]	0.08 [1.56]	0.07 [1.59]	0.11 [2.18]	0.29 [4.11]
α_{FF5}	-0.19 [-3.52]	-0.06 [-1.27]	0.09 [1.73]	0.04 [0.87]	0.06 [1.08]	0.25 [3.64]
α_{FF6}	-0.18 [-3.17]	-0.04 [-0.82]	0.12 [2.16]	0.02 [0.36]	0.05 [1.01]	0.23 [3.31]
Panel B: Fama and French (2018) 6-factor model loadings for DCU-sorted portfolios						
β_{MKT}	1.09 [84.23]	0.98 [94.12]	0.97 [76.54]	0.97 [91.61]	1.04 [86.56]	-0.05 [-3.28]
β_{SMB}	0.10 [5.35]	-0.11 [-6.99]	-0.02 [-0.92]	-0.04 [-2.37]	0.13 [7.27]	0.03 [1.11]
β_{HML}	0.07 [2.93]	0.14 [7.10]	-0.14 [-5.96]	-0.02 [-0.91]	-0.03 [-1.50]	-0.11 [-3.48]
β_{RMW}	0.09 [3.52]	0.12 [5.81]	-0.04 [-1.80]	0.07 [3.45]	0.10 [4.42]	0.01 [0.46]
β_{CMA}	-0.12 [-3.30]	0.03 [1.11]	-0.10 [-2.61]	0.16 [5.21]	0.15 [4.19]	0.27 [5.79]
β_{UMD}	-0.03 [-2.23]	-0.03 [-3.08]	-0.04 [-3.01]	0.04 [3.55]	0.00 [0.42]	0.03 [2.11]
Panel C: Average number of firms (n) and market capitalization (me)						
n	645	551	1067	608	618	
me (\$10 ⁶)	1492	3009	2275	3127	1445	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCU strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27 [3.90]	0.32 [4.77]	0.32 [4.71]	0.29 [4.11]	0.25 [3.64]	0.23 [3.31]
Quintile	NYSE	EW	0.24 [5.10]	0.26 [5.65]	0.25 [5.37]	0.23 [4.82]	0.21 [4.47]	0.20 [4.17]
Quintile	Name	VW	0.25 [3.73]	0.32 [4.81]	0.32 [4.75]	0.27 [4.04]	0.26 [3.85]	0.23 [3.40]
Quintile	Cap	VW	0.23 [3.57]	0.27 [4.26]	0.28 [4.30]	0.22 [3.43]	0.19 [2.95]	0.15 [2.41]
Decile	NYSE	VW	0.30 [3.07]	0.37 [3.91]	0.38 [3.94]	0.35 [3.54]	0.30 [3.07]	0.28 [2.88]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [3.05]	0.26 [3.79]	0.26 [3.76]	0.24 [3.45]	0.20 [2.88]	0.18 [2.64]
Quintile	NYSE	EW	-0.02 [-0.30]					
Quintile	Name	VW	0.20 [2.87]	0.25 [3.86]	0.25 [3.77]	0.22 [3.41]	0.20 [3.01]	0.18 [2.74]
Quintile	Cap	VW	0.18 [2.78]	0.23 [3.52]	0.23 [3.53]	0.20 [3.08]	0.16 [2.44]	0.13 [2.09]
Decile	NYSE	VW	0.23 [2.35]	0.30 [3.13]	0.30 [3.15]	0.29 [2.95]	0.24 [2.44]	0.22 [2.28]

Table 3: Conditional sort on size and DCU

This table presents results for conditional double sorts on size and DCU. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCU. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCU and short stocks with low DCU. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCU Quintiles					DCU Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.68 [2.44]	0.88 [3.22]	0.92 [3.39]	0.90 [3.23]	0.79 [2.85]	0.11 [1.22]	0.15 [1.63]	0.14 [1.54]	0.08 [0.91]	0.07 [0.76]	0.04 [0.39]
	(2)	0.72 [2.68]	0.94 [3.71]	0.82 [3.25]	0.83 [3.39]	0.89 [3.50]	0.18 [2.30]	0.21 [2.65]	0.17 [2.27]	0.17 [2.26]	0.12 [1.56]	0.13 [1.63]
	(3)	0.79 [3.14]	0.78 [3.52]	0.85 [3.52]	0.79 [3.52]	0.91 [3.87]	0.12 [1.58]	0.17 [2.25]	0.17 [2.20]	0.12 [1.56]	0.14 [1.79]	0.11 [1.38]
	(4)	0.70 [3.02]	0.80 [3.87]	0.85 [3.81]	0.73 [3.51]	0.91 [4.19]	0.21 [2.82]	0.26 [3.45]	0.24 [3.23]	0.20 [2.72]	0.21 [2.75]	0.18 [2.39]
	(5)	0.45 [2.26]	0.62 [3.55]	0.60 [3.03]	0.66 [3.64]	0.78 [4.07]	0.32 [3.74]	0.36 [4.15]	0.38 [4.32]	0.31 [3.57]	0.30 [3.38]	0.26 [2.91]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCU Quintiles					DCU Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	390	393	392	393	388	37	34	32	33	35	
	(2)	107	107	107	107	107	61	61	60	61	61	
	(3)	76	77	76	76	76	107	109	104	106	108	
	(4)	64	65	64	65	64	231	238	230	232	231	
(5)	59	59	59	59	59	1387	2114	1956	2138	1580		

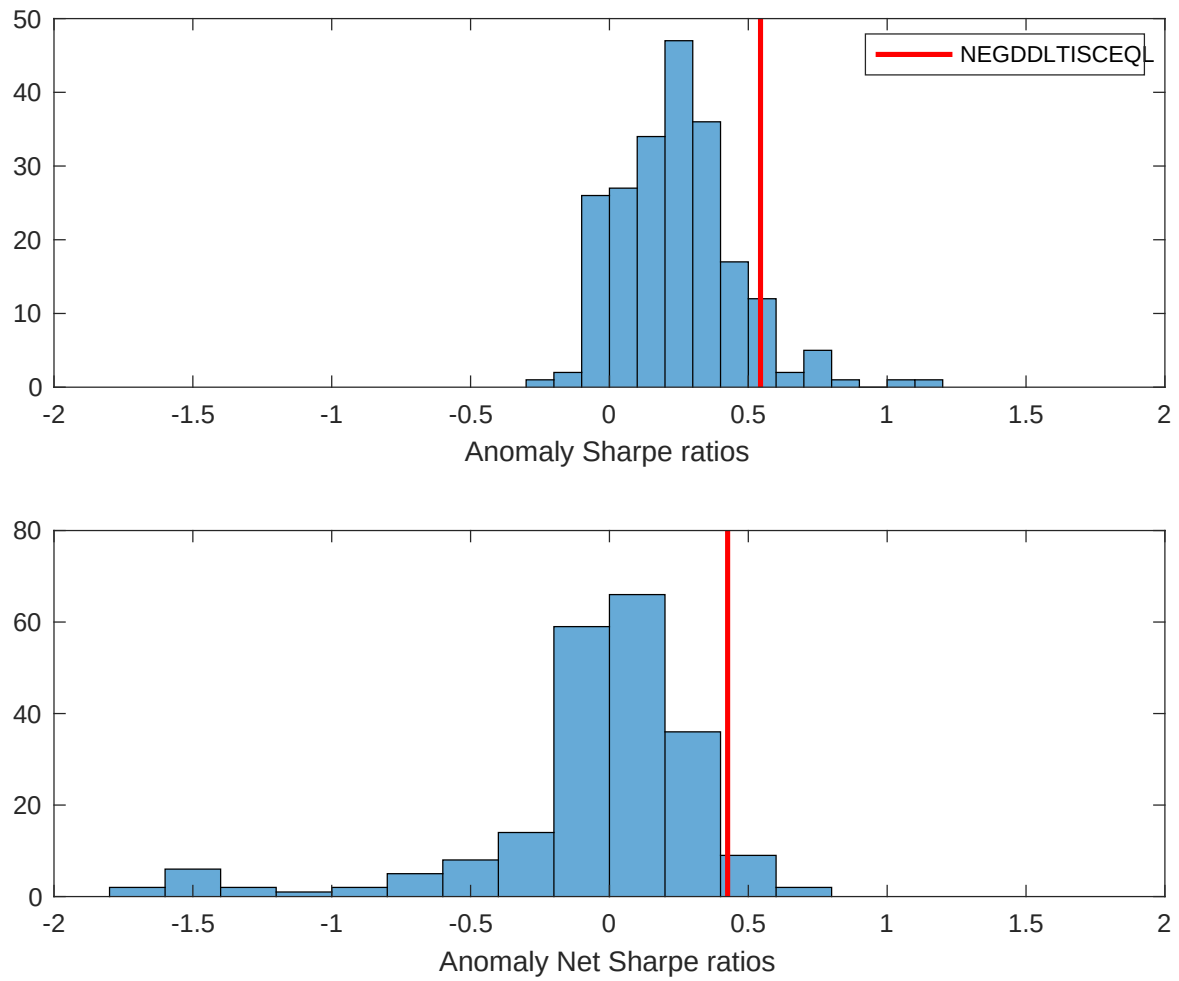


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCU with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

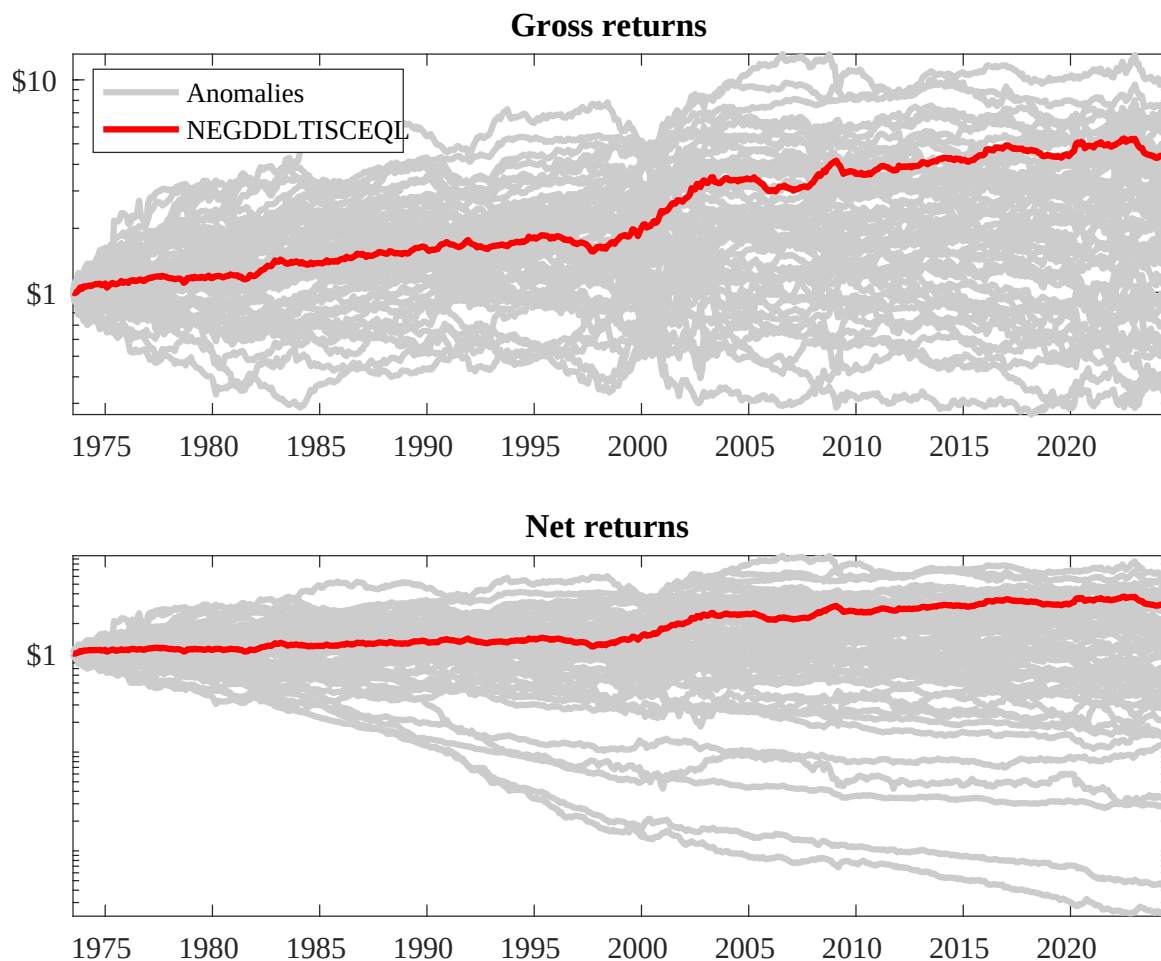
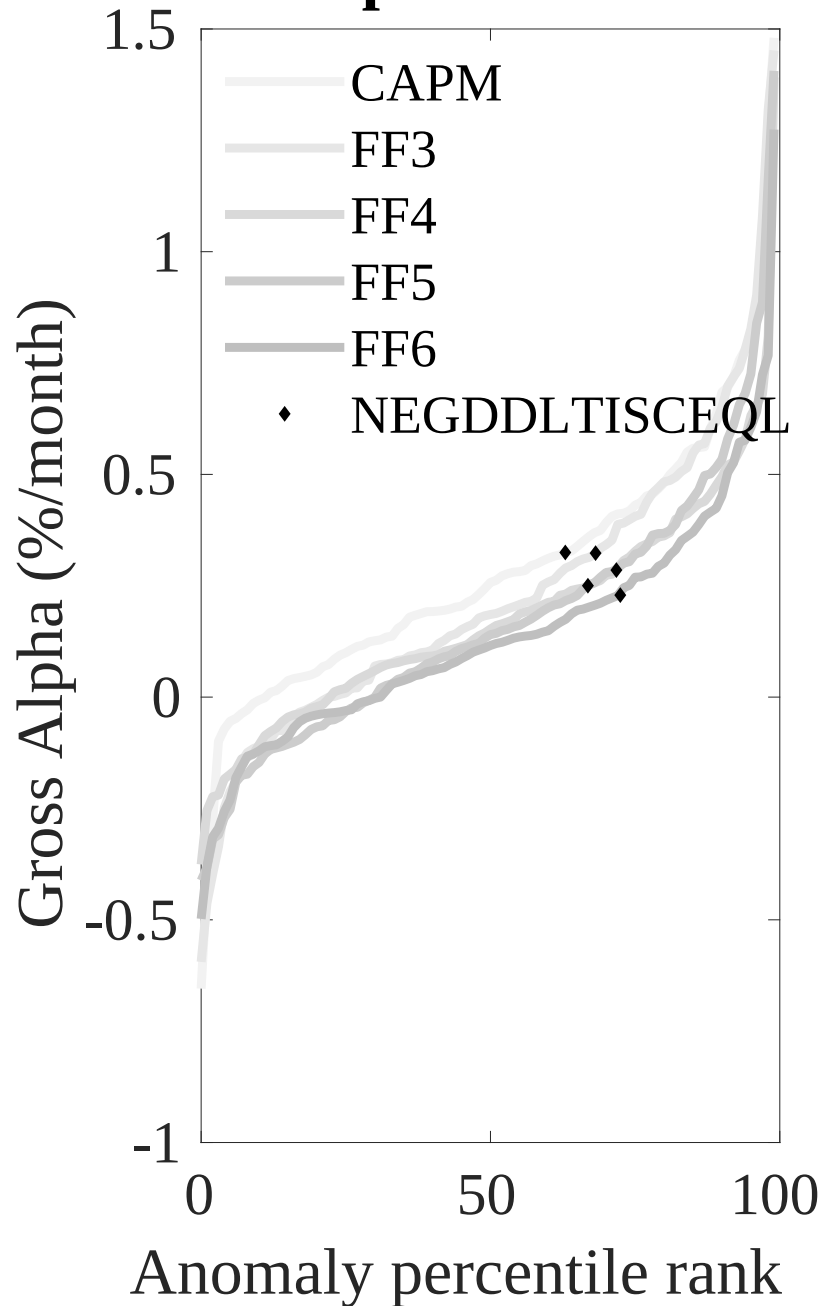


Figure 3: Dollar invested.

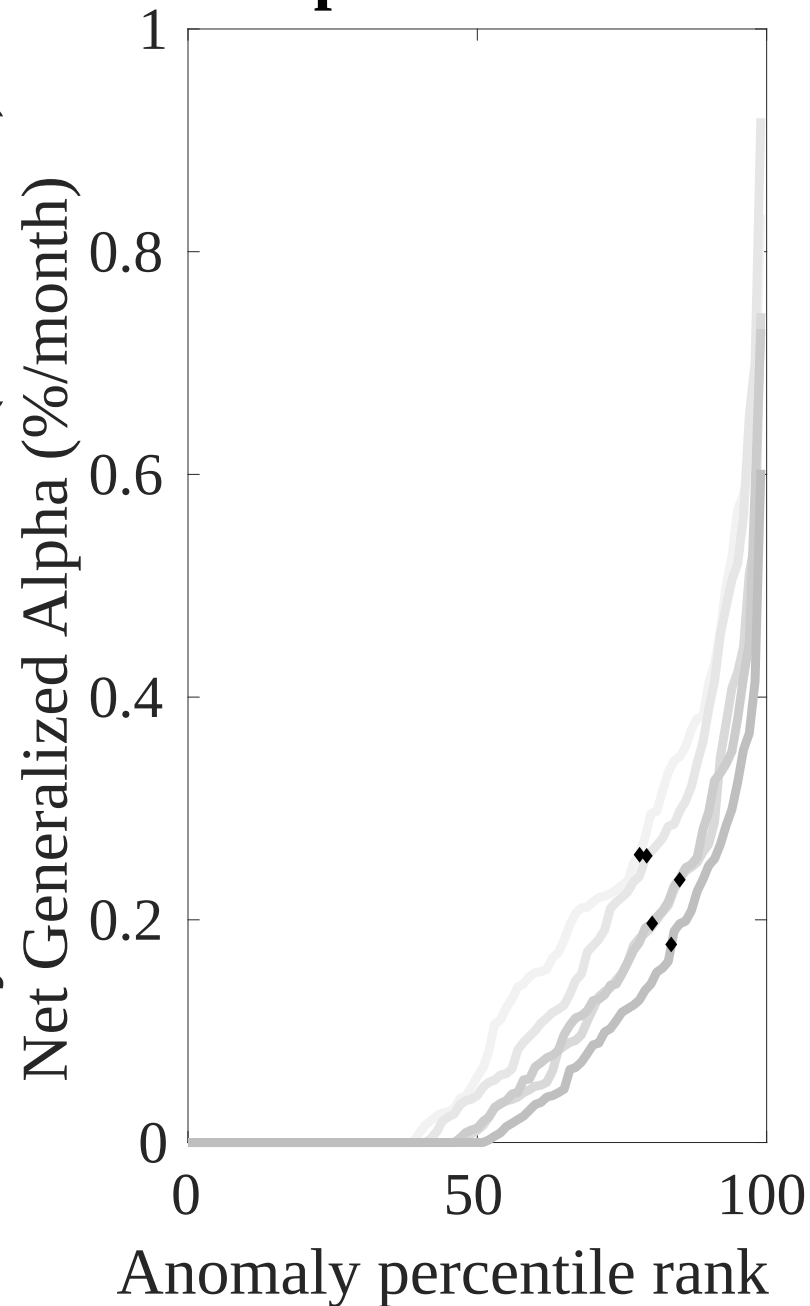
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCU trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (RFS, 2016)



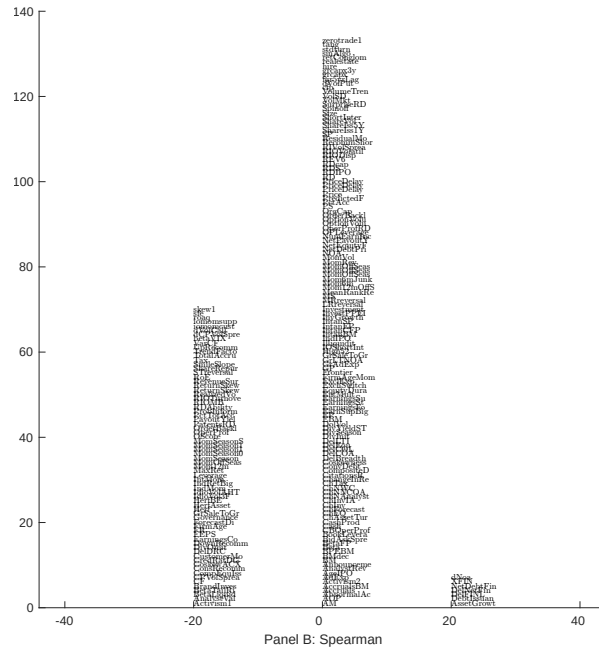
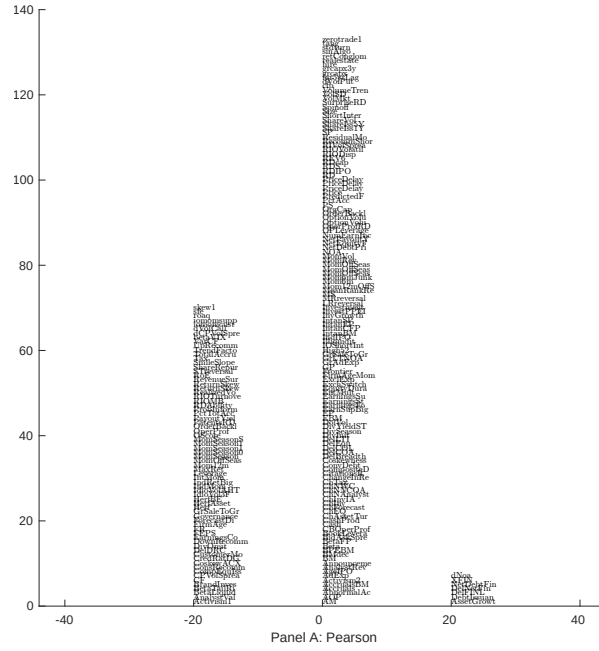


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with DCU. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

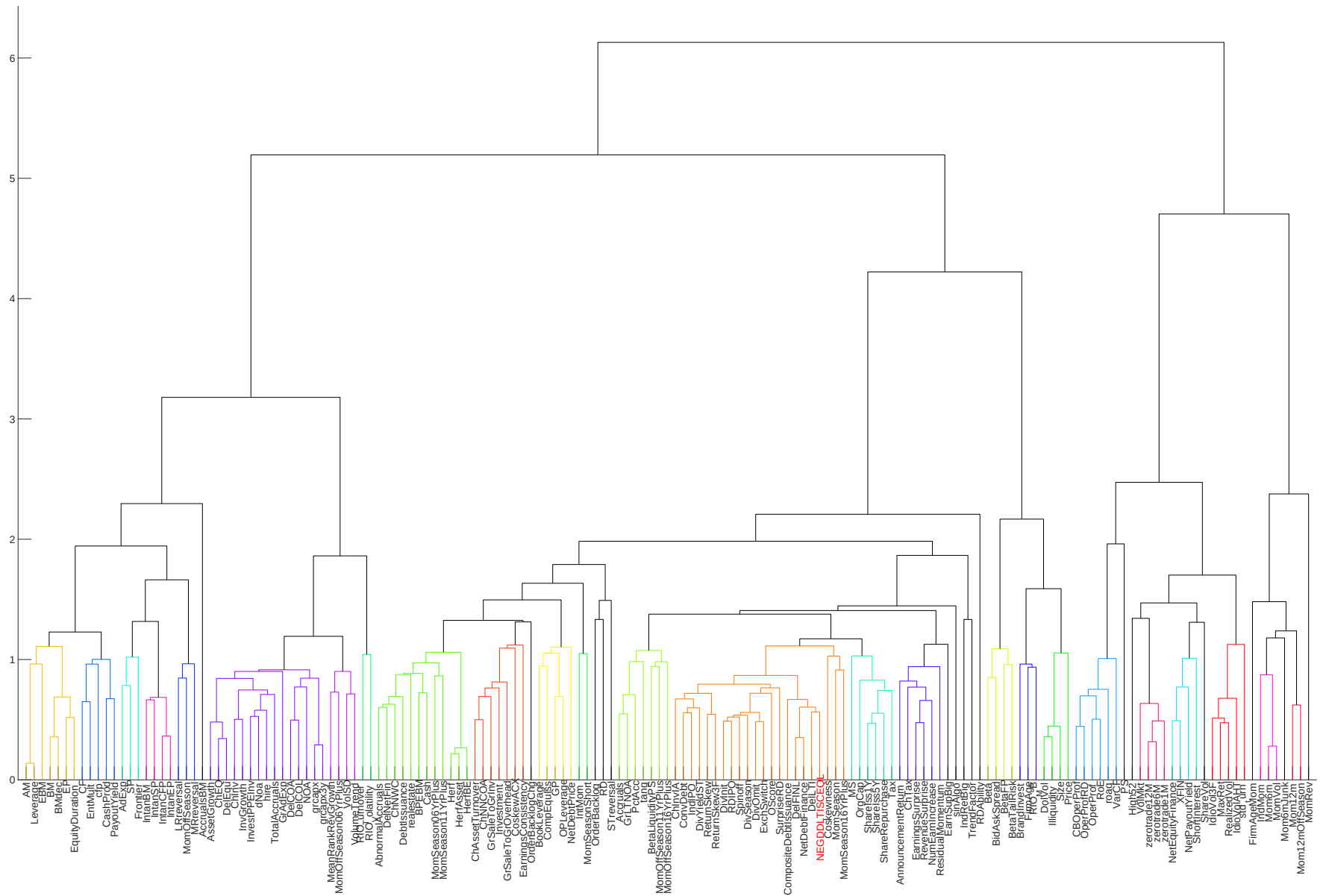


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

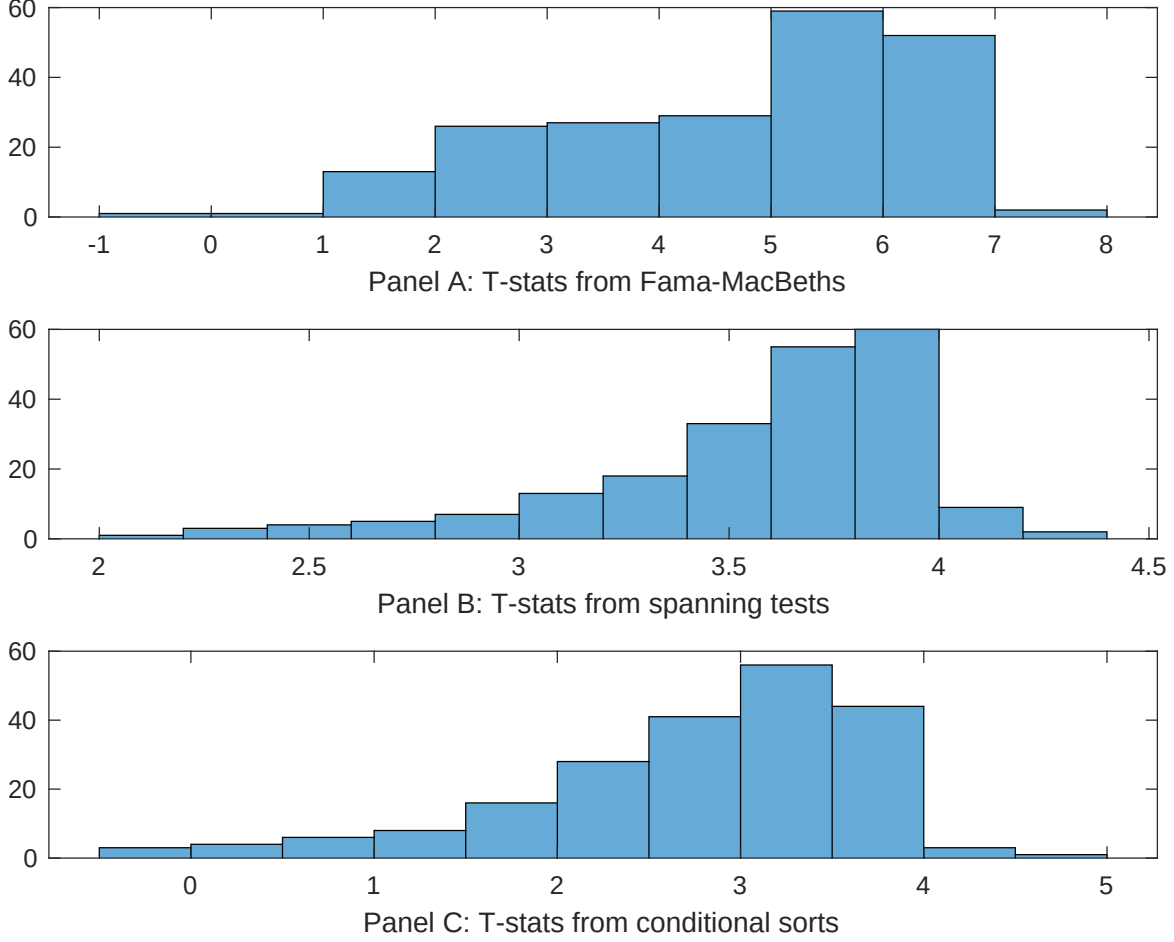


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCU conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCU} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCU,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCU. Stocks are finally grouped into five DCU portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCU trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCU. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCU}DCU_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.13 [5.32]	0.14 [5.69]	0.14 [5.80]	0.14 [5.30]	0.14 [5.66]	0.15 [5.70]
DCU	0.48 [1.98]	0.56 [2.22]	0.73 [2.75]	0.38 [1.52]	0.11 [4.22]	0.34 [1.44]	0.79 [0.27]
Anomaly 1	0.17 [9.27]						-0.11 [-2.30]
Anomaly 2		0.20 [8.82]					0.99 [1.50]
Anomaly 3			0.19 [6.37]				0.98 [1.68]
Anomaly 4				0.11 [8.69]			0.41 [1.80]
Anomaly 5					0.42 [7.14]		0.43 [0.78]
Anomaly 6						0.14 [9.88]	0.95 [5.25]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCU trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCU} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.20 [2.95]	0.20 [3.01]	0.20 [2.95]	0.21 [3.15]	0.22 [3.25]	0.21 [3.07]	0.20 [2.98]
Anomaly 1	16.33 [4.27]						10.30 [1.96]
Anomaly 2		18.42 [5.02]					9.53 [1.90]
Anomaly 3			13.91 [4.11]				10.20 [2.82]
Anomaly 4				7.42 [1.87]			2.85 [0.68]
Anomaly 5					6.89 [2.60]		6.46 [2.40]
Anomaly 6						3.96 [1.01]	-6.41 [-1.47]
mkt	-4.72 [-3.04]	-5.06 [-3.28]	-3.19 [-1.96]	-5.10 [-3.25]	-5.22 [-3.33]	-5.08 [-3.23]	-3.57 [-2.21]
smb	-0.28 [-0.12]	-0.23 [-0.10]	5.86 [2.26]	0.86 [0.36]	2.15 [0.90]	1.47 [0.62]	3.16 [1.16]
hml	-10.56 [-3.52]	-10.90 [-3.66]	-10.10 [-3.35]	-11.98 [-3.95]	-11.86 [-3.92]	-11.97 [-3.92]	-9.26 [-3.06]
rmw	-0.17 [-0.06]	-0.73 [-0.24]	-7.01 [-1.92]	1.33 [0.43]	2.30 [0.74]	1.36 [0.44]	-5.93 [-1.61]
cma	23.30 [4.94]	23.99 [5.19]	19.70 [3.90]	19.79 [2.98]	22.64 [4.39]	25.83 [4.76]	11.76 [1.74]
umd	2.10 [1.33]	2.32 [1.50]	3.41 [2.21]	3.67 [2.35]	2.81 [1.77]	3.38 [2.15]	1.53 [0.97]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	14	15	14	12	13	12	17

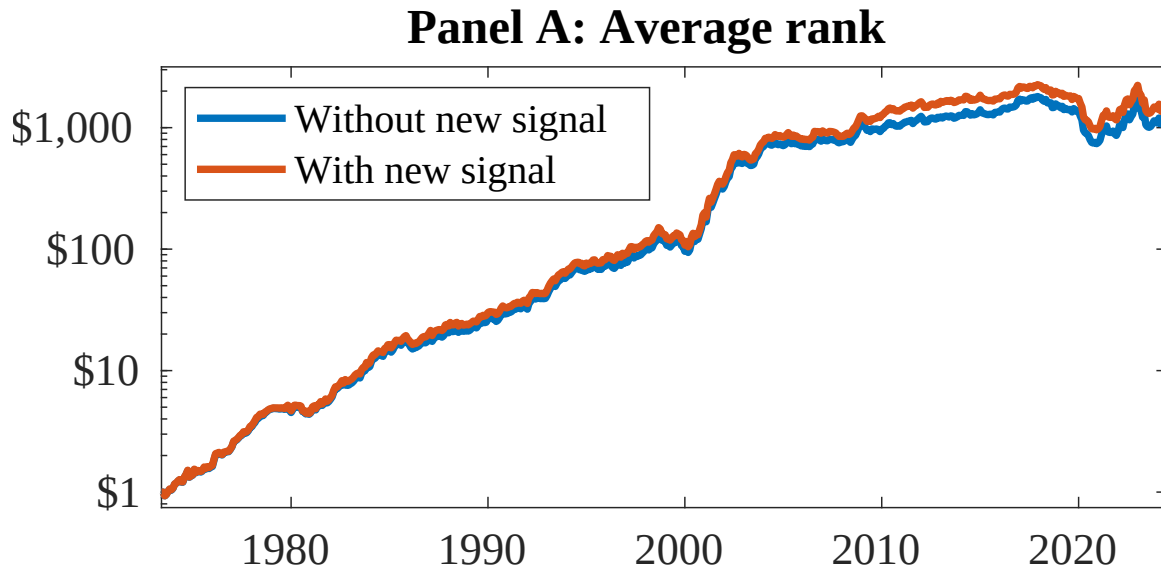


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DCU. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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